Norris Phase Report 4

Main devtest execution under the frozen protocol

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Repository github.com/SHAZAN01/csudh-norris-ape-2026, branch phase4-main

Phase 4 complete: 24 conditions × 1,012 items = 24,288 post-edited outputs, plus the S0 baseline. Generated only; nothing has been scored.

Executive summary

Phase 4 ran on 24 October 2026 between 07:33 and 09:21 UTC under protocol freeze 6e6259a9cf1c…, declared that morning. The sentence-split NLLB baseline (S0) was generated for FLORES+ devtest, the 12,144 frozen jobs were prepared, and both editors completed their full matrices after smoke runs. Every planned output exists, aligned to its 1,012 devtest IDs per direction.

Output validity was complete. All 12,144 Llama 3 8B and all 12,144 GPT-4o Mini outputs parsed, matched the schema, ended normally and were accepted: **0 baseline fallbacks in all 24 conditions**, including both reason-then-edit conditions. This matches the development gate (0/400 per editor) and means Phase 5 measures editing behaviour directly rather than a mixture with fallback.

Execution integrity. GPT-4o Mini sent exactly 12,144 requests, all returning HTTP 200, with 0 retries and no uncertain or repeated paid calls. Llama ran 762 vLLM batches without failure. The code that executed is byte-identical to the frozen code at every stage. Regenerated unsplit devtest drafts match the Phase 2 archive on 4,048/4,048 items. No devtest output was scored or inspected for quality in this phase.

Approved-plan acceptance condition	Evidence
Frozen provenance on every record	All records validated against run hashes; run specs bind freeze and protocol fingerprint
Executed code equals frozen code	Engine, protocol module, job preparer, splitter and S0 generator hashes identical to the freeze
Exactly 24 conditions × 1,012 unique IDs	24/24 condition files; 1,012 unique IDs each; 24,288 outputs
Aligned sources	Every record matched to its devtest source; S0 segmentation lossless
Non-empty translation for every item	All 24,288 accepted; 0 fallbacks (fallback path tested, unused)
No unresolved transport states	0 unresolved attempt markers; 12,144/12,144 HTTP 200; 0 retries
Files reopened and checksummed	Each condition file reopened and SHA-256 recorded before completion
References never in prompts	12,144 prompts checked: 0 leaks
Nothing scored before analysis	scored = false throughout Phase 4

1. Frozen provenance

Item	Value
Protocol freeze	6e6259a9cf1c7e2aca978a770fa85d86… declared 2026-10-24T06:58:38 UTC; commit 51345a22bc21
Protocol fingerprint	7547a429e81ae64bd8a5834903eb3492… (identical in both editor run specs)
Prepared jobs	main_adb98f4e942e213b; manifest c5d7a062334cba62…; preparer 8cf61c83aa1b3574… = frozen
Engine	run_main_experiments.py eec6dab29d9b4bc5… = frozen; protocol module = frozen
S0 run	ac593778a8509b9a7a0c6fa3…; splitter and generator = frozen; variant raw_len_chars_desc_b32
Llama run	1c9083d0022e90d1…; vLLM 0.11.2, torch 2.9.0, Transformers 4.57.1; BF16; revision 8afb486c1db2
GPT run	eb319d52dbcbe4fe…; gpt-4o-mini-2024-07-18; openai 1.109.1 (AsyncOpenAI, max_retries 0); 4 req/s, concurrency 4

2. Primary baseline (S0) on devtest

Single-sentence items reuse the checksum-locked Phase 2 unsplit drafts (as on dev); multi-sentence items are split by the frozen splitter and their fragments translated with the frozen variant (raw fragments, length-descending, batch 32). The unsplit control was also regenerated and compared with the archive.

Direction	Multi-sentence items	Fragments translated	Fragment GPU time (s)	Split ≠ unsplit	Unsplit regenerated = archive
En→Tl	113	230	8.2	105	1012/1,012
Tl→En	118	240	6.6	104	1012/1,012
En→Es	113	230	6.5	98	1012/1,012
Es→En	122	252	7.2	114	1012/1,012

The two English-source directions share one segmentation (113 multi-sentence items, 230 fragments each), as expected. Between 98 and 114 items per direction differ from the unsplit draft. The exact regeneration of all 4,048 unsplit drafts confirms that the generation environment reproduces the Phase 2 run. The S0 dev-reproduction disposition (412/421 multi-sentence dev items) remains with you for acknowledgement (Report 3.2, Section 5).

3. Retrieval and job preparation

LaBSE (836121a0533e) embedded all 997 dev sources per source language; the embeddings reproduce the pilot's stored matrices exactly (maximum absolute difference 0.0 for English, Tagalog and Spanish). Each devtest query retrieved its top-1 dev exemplar by exact FAISS inner product with smallest-ID tie break; no devtest query had an exact-duplicate dev source to exclude. The preparer built 12,144 jobs (1,012 IDs × 4 directions × 3 strategies) shared by both editors, and the reference-leak check found 0 of 12,144 prompts containing their own devtest reference.

4. Smoke stage

Editor	Outputs	Parsed	Schema-valid	Usable	Fallback	Median latency
GPT-4o Mini	24	24	24	24	0	1070 ms (request)
Llama 3 8B	24	24	24	24	0	335 ms (batch-amortized)

Smoke IDs [0, 23] × 4 directions × 3 strategies per editor; these records were reused by the full runs (identical run specifications), not regenerated.

5. Execution accounting per condition

Editor	Strategy	Dir.	n	Usable	FB	Prompt tok.	Compl. tok.	Median ms
GPT-4o Mini	Zero-shot	En→Tl	1012	1012	0	342,496	48,495	976
GPT-4o Mini	Zero-shot	Tl→En	1012	1012	0	344,416	32,055	816
GPT-4o Mini	Zero-shot	En→Es	1012	1012	0	331,265	38,200	871
GPT-4o Mini	Zero-shot	Es→En	1012	1012	0	334,897	33,310	838
GPT-4o Mini	One-shot	En→Tl	1012	1012	0	507,223	48,467	978
GPT-4o Mini	One-shot	Tl→En	1012	1012	0	493,114	32,268	824
GPT-4o Mini	One-shot	En→Es	1012	1012	0	474,970	38,361	888
GPT-4o Mini	One-shot	Es→En	1012	1012	0	475,653	33,535	840
GPT-4o Mini	Reason-then-edit	En→Tl	1012	1012	0	458,876	73,286	1216
GPT-4o Mini	Reason-then-edit	Tl→En	1012	1012	0	460,796	55,792	1034
GPT-4o Mini	Reason-then-edit	En→Es	1012	1012	0	447,645	58,114	1092
GPT-4o Mini	Reason-then-edit	Es→En	1012	1012	0	451,277	53,780	1030
Llama 3 8B	Zero-shot	En→Tl	1012	1012	0	311,958	60,013	140
Llama 3 8B	Zero-shot	Tl→En	1012	1012	0	313,999	35,408	80
Llama 3 8B	Zero-shot	En→Es	1012	1012	0	297,225	47,162	101
Llama 3 8B	Zero-shot	Es→En	1012	1012	0	301,200	36,750	84
Llama 3 8B	One-shot	En→Tl	1012	1012	0	486,348	59,861	143
Llama 3 8B	One-shot	Tl→En	1012	1012	0	463,237	34,369	83
Llama 3 8B	One-shot	En→Es	1012	1012	0	444,175	46,406	108
Llama 3 8B	One-shot	Es→En	1012	1012	0	439,215	35,635	87
Llama 3 8B	Reason-then-edit	En→Tl	1012	1012	0	414,170	103,616	239
Llama 3 8B	Reason-then-edit	Tl→En	1012	1012	0	416,211	76,254	172
Llama 3 8B	Reason-then-edit	En→Es	1012	1012	0	399,437	80,045	177
Llama 3 8B	Reason-then-edit	Es→En	1012	1012	0	403,412	72,368	159

Median ms is HTTP request wall time for GPT-4o Mini and batch-amortized generation time for Llama (16 sequences per batch, initialization excluded); the two are not comparable per-item speeds. All 24 conditions: JSON parse, schema and usable counts are 1,012/1,012, and the only failure class present is "usable".

6. Totals, runtime and cost

Measure	Llama 3 8B Instruct	GPT-4o Mini
Outputs / usable / fallback	12,144 / 12,144 / 0	12,144 / 12,144 / 0
Prompt / completion tokens	4,690,587 / 687,887	5,122,628 / 545,663
Retries	0	0 (12,144 requests, all HTTP 200)
Wall time	762 batches; 1,634.8 s summed batch time; full invocation 07:49–08:21 UTC	Four direction invocations, 08:21–09:21 UTC (about 15 min each at 4 req/s)
Server variation	—	8 system_fingerprint values within the dated snapshot
Estimated cost	Local A100 (no API cost)	$1.10 at list uncached token rates (not an invoice)

Descriptive generation features (not quality): retrieved one-shot prompts are about 1.4–1.6 times longer than zero-shot prompts, and reason-then-edit completions are about 1.5–2.2 times longer than zero-shot completions because they include the diagnosis. Whether and how the editors changed the drafts, and with what effect, is a Phase 5 question.

7. Incidents and deviations

• **No protocol deviations.** No prompt, model, decoding, retrieval, contract or code change occurred after the freeze.
• Daily API cap: GPT-4o Mini completed all 12,144 requests in one session; no rate-limit response occurred.
• Evidence packaging: the Phase 4 ZIP stored files with repeated base names (three run_spec.json files, for example), which stalls standard unzip. All entries were recovered by index and verified; the notebook ZIP cell now uses unique paths. This affects packaging only, not results.
• A benign PyTorch NCCL shutdown warning appears at the end of each vLLM process.
• Raw generations remain private on Drive (benchmark text); this report and the repository contain numeric summaries only.

8. Next steps

Phase 5 now scores the frozen matrix exactly as declared: COMET-22 (primary), BLEU and chrF++ with signatures; F1 (24 conditions vs S0) and F2 (24 strategy contrasts) with centred paired bootstrap and Holm correction; improvement/harm bands; protect-pool over-editing with sensitivity thresholds; preservation screens; and source review of flagged cases. The statistical procedure passed its synthetic validation fixtures before any devtest score exists. Report 5 and the held-out manuscript sections follow, keeping the 24 November full-draft date.

Evidence

Phase 4 evidence ZIP SHA-256 66b8d20ed6f480bb0bea4e8b40de09820ad436cdc00ba579fda14f6aa33677aa
Protocol freeze 6e6259a9cf1c7e2aca978a770fa85d86596f31b0e8b3c55f42635203e4f7462c
S0 run ac593778a8509b9a7a0c6fa3923ae685671bdfe35a3f515050219d8b7fa60106
Llama run 1c9083d0022e90d1ff6868313cf221013b3c8fc91e9fc3e0959ecdd9906b3468
GPT run eb319d52dbcbe4fe95b5d2acb76189994e93f780e6bde1f8a3de58ab771b0e27